Digital Signal Processing

Signal Categorization
A/D and D/A processes

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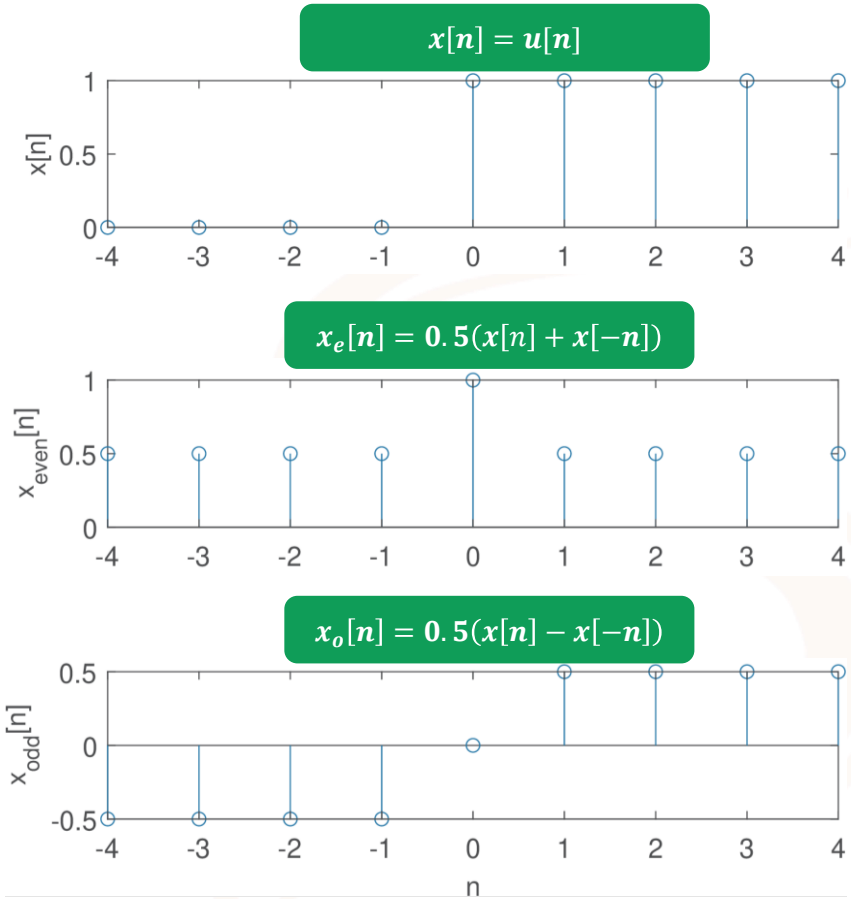
Even and odd signals

A signal $x[n]$ is called *even* if
$x[n] = x[-n]$ *

A signal $x[n]$ is called *odd* if
$x[n] = -x[-n]$ *

Any signal can be written as a sum of an even and an odd parcel:
$x[n] = x_e[n] + x_o[n]$



* Definitions are also valid for signals $x(t)$, in general for any function $f(x)$.

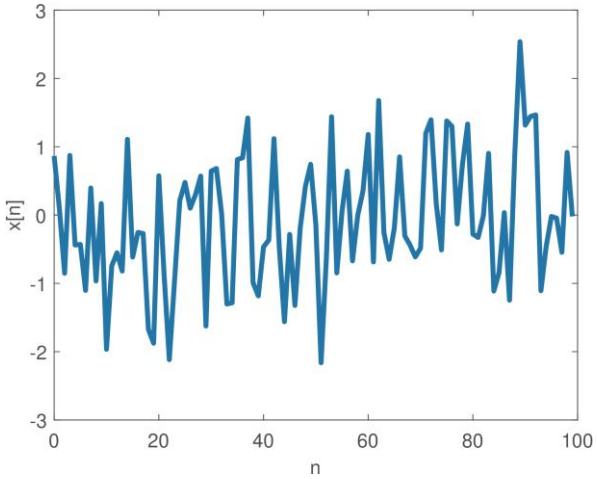
Random signals and their generation

Random signals are important to represent noise or other non-deterministic signals

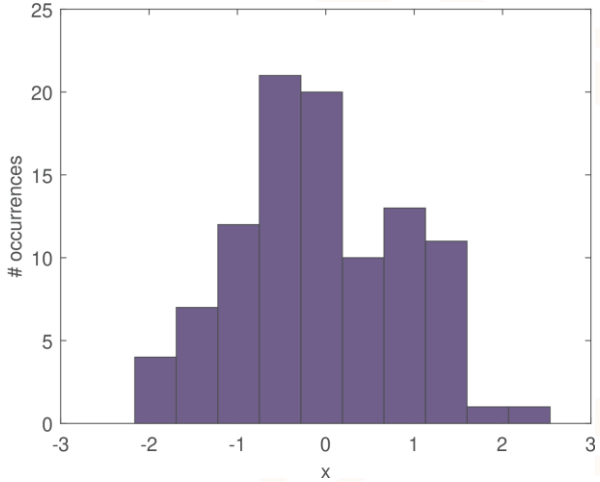
Discrete-time random signals of finite duration can be represented by vectors in which the elements are outcomes of random variables

It is simple to generate Random signals in Matlab/Octave

`x=randn(1,100)`
Generates 1x100 samples with $x[n] \sim \mathcal{N}(0,1)$



Waveform



Histogram

$\mathcal{N}(\mu, \sigma^2)$

numpy.random generates normal as `normal(`μ, σ`)`

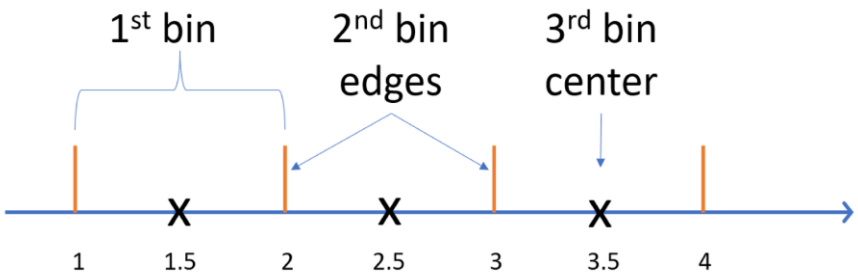
Histogram calculation

In Python, you can generate a histogram using the code:

```
import numpy as np
x=np.array([1, 3.1, 2.6, 4])
hist, bin_edges = np.histogram(x,3)
```

To calculate the bin width, the following formula is used:

$$\text{bin_width} = \frac{x_{\max} - x_{\min}}{B}$$



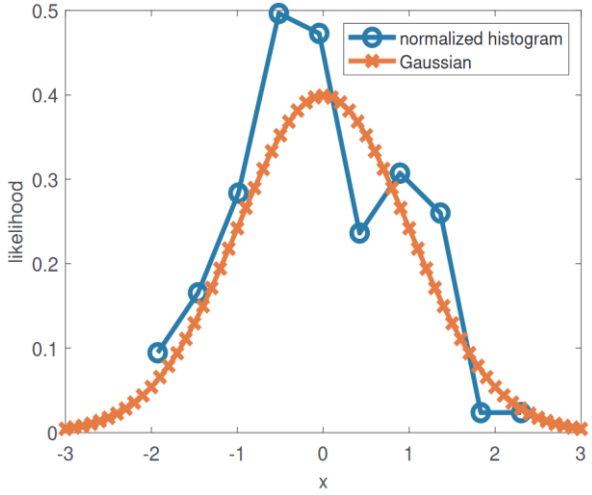
Histogram generated by the code, the 'x' marks the bin centers

PMF and PDF via histogram

Firstly, to obtain a estimation of the PMF and PDF, one assumption needs to be considered: the bin centers must be interpreted as the output of a discrete random variable.

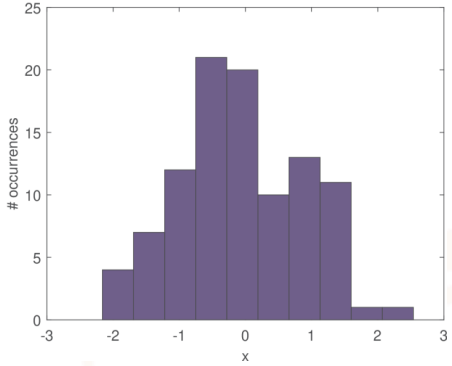
Estimated PMF: divide the bin centers by the total number of samples in the histogram

Estimated PDF: divide the estimated PMF result by the bin width



Example of PDF x Estimated PDF

Estimated from:



Changing the mean and variance of a random variable

X is a random variable that has mean $\mathbb{E}[X] = \eta$ and variance σ_x

x is a vector composed by N realizations of X

Changing the mean:
New Y with $\mathbb{E}[Y] = \eta + \kappa$

$Y = X + \kappa$	$y = x + \kappa$

Changing the variance:
New Y with $\sigma_Y^2 = \kappa\sigma_X^2$

$Y = X\sqrt{\kappa}$	$y = x\sqrt{\kappa}$

If you have a Gaussian $\mathcal{N}(0,1) = X$, you can make $Y = \sigma X + \eta$ to obtain a new random variable with mean η and σ^2 variance. It is specially useful when using Python or Matlab in codes like this:

```
newMean=4; %new mean
newVariance=0.09; %new variance
N=10000; %number of random samples
x=sqrt(newVariance)*randn(1,N)+newMean;
```

Proofs: Let X be a random variable with mean $\mathbb{E}[X] = \eta$ and variance σ_X

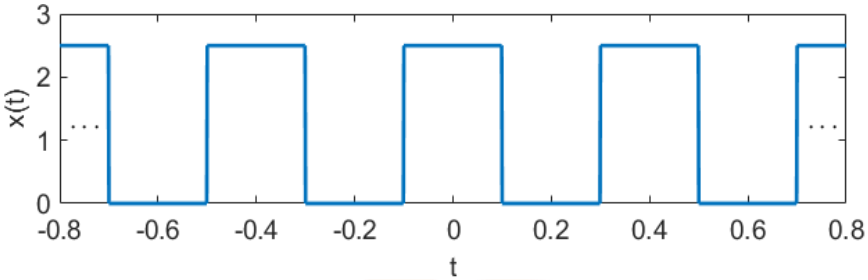
Changing the mean: $\mathbb{E}[Y] = \mathbb{E}[X + \kappa] = \mathbb{E}[X] + \mathbb{E}[\kappa] = \eta + \kappa$

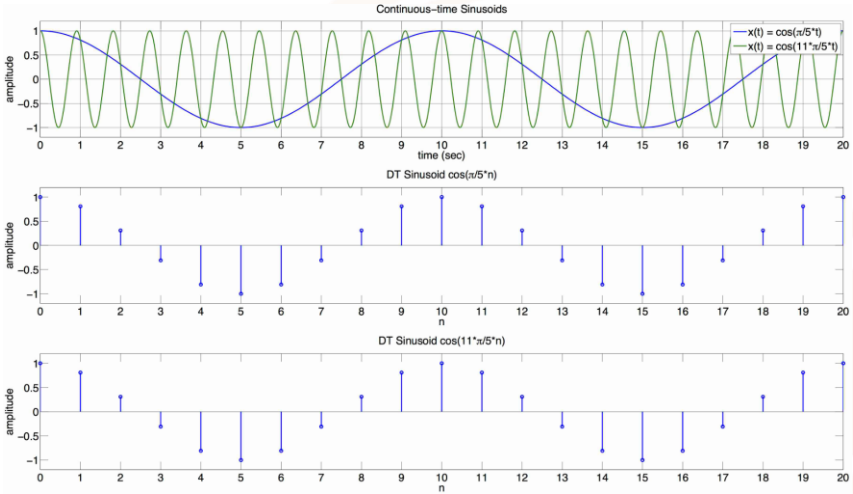
Changing the variance: $\sigma_y^2 = \mathbb{E}[(Y - \mathbb{E}[Y])^2] = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 = \mathbb{E}\left[\left(\sqrt{\kappa}X\right)^2\right] - \mathbb{E}[\sqrt{\kappa}X]^2 = \kappa\left(\mathbb{E}[Y^2] - \mathbb{E}[Y]^2\right) = \kappa\sigma_Y^2$

Periodic and aperiodic Signals

A signal $x(t)$ is periodic if a segment of $x(t)$ is eternally repeated, such that:
$x(t) = x(t+T)$, for some $T > 0$

If the T is the smallest value possible that $x(t) = x(t+T)$, we call it the *fundamental period* T_0





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The same discussion is also valid for signals $x[n]$ (but instead of period $T \in \mathbb{R}: T > 0$, the period is $N \in \mathbb{Z}, \mathrm{N} > 0$)

LCM and GCD for a periodic signal composed by commensurate frequencies

If a signal is composed of two or more frequencies and they are commensurate, i.e., they can be written as a ratio between two integers, we can use the Least Common Multiple (LCM) to get the *fundamental period* T_0 or the Greatest Common Divisor (GCD) to get the *fundamental frequency* F_0

For example:

$$x(t) = \cos\left(2\pi\frac{5}{2}t\right) + \cos\left(2\pi\frac{1}{6}t + \frac{\pi}{2}\right) + \sin\left(2\pi\frac{1}{8}t\right)$$

$F_1 = \frac{5}{2}\text{Hz}, T_1 = 0.4\text{ s}$

$F_2 = \frac{1}{6}\text{Hz}, T_1 = 6\text{ s}$

$F_1 = \frac{1}{8}\text{Hz}, T_1 = 8\text{ s}$

To find the LCM, we need to multiply by 10 to eliminate the decimal part, resulting in LCM(4, 60, 80) = 240. Then, we divide by 10 and obtain $T_0 = 24$ s

If a signal is composed of incommensurate frequencies (example: 1Hz and π Hz), then it is not a periodic signal

Pay attention: Sinusoids in discrete-time are not always periodic!

A discrete-time sinusoid such as $x[n] = A\cos(\Omega n + \phi)$ is periodic if and only if:

$\frac{\Omega}{2\pi}$ is a ratio between integers in irreducible form $\frac{m}{N_0}$

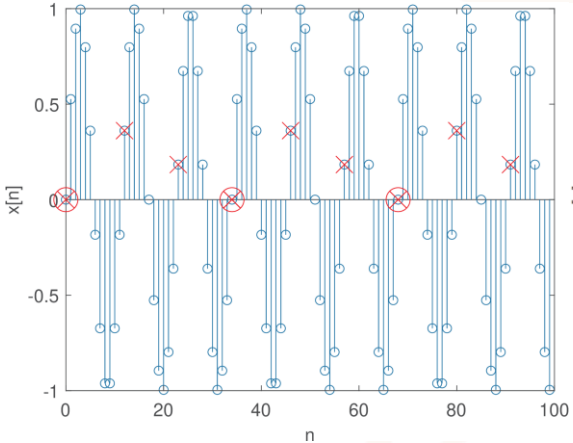
N_0 is the fundamental period
m is the number of sine envelopes within N_0 samples

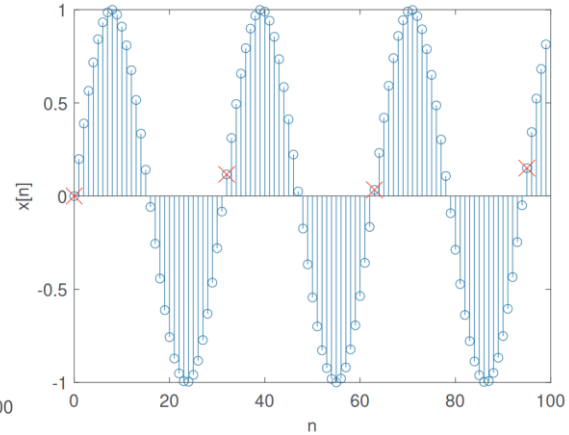
$x[n] = \sin\left(\frac{3\pi}{17}\right)n$
Periodic!

It repeats every 34 samples

$x[n] = \sin(0.2n)$
Not Periodic!

The crosses will never repeat!





The crosses indicate the start of a new sine envelope

Power and Energy Signals

Instantaneous power

$$p(t) = |x(t)|^2$$

Energy

$$E = \int_{-\infty}^{\infty} p(t)\mathrm{d}t$$

Average Power

$$\mathcal{P} = \lim_{\Delta t \to \infty} \left[\frac{1}{\Delta t} \int_{-\Delta t/2}^{\Delta t/2} p(t)\mathrm{d}t \right]$$

It is possible to deduce the signal categories by assigning infinite or finite values in the power formula and intuitively observing the result

Type of Signals

Category	E	$\mathcal{P}$	Example(s)
Power signal	∞	finite	$\cos(\omega t)$ and other periodic signals
Energy signal	finite	0	$e^{-t}u(t)$ and $t^2[u(t) - u(t-5)]$
Neither	∞	∞	t

Quiz-Time!

Periodicity when…?

Random Noise?

Odd or Even?

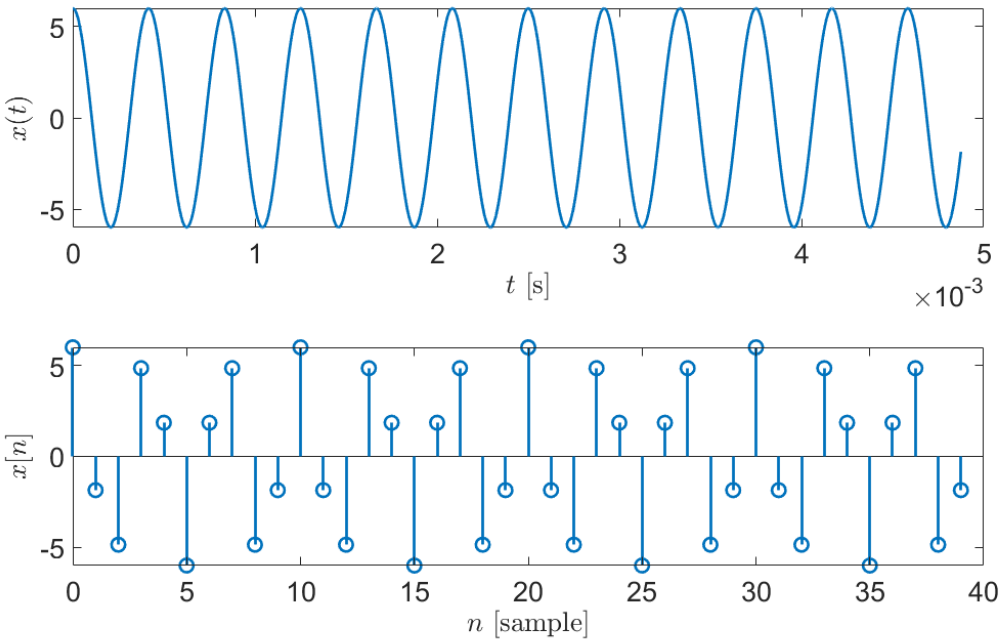
Power or Energy?

How to discretize and reconstruct a signal?

Signals in nature are analog, that is, infinite information within a time interval

However, under some (ideal) conditions, we are able to mathematically represent an analog signal as a discrete-time sequence without loss of information and then reconstruct it back as analog

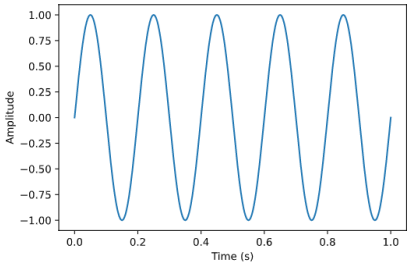
We then proceed to describe the required steps for discretization and reconstruction



Analog to digital (A/D) conversion

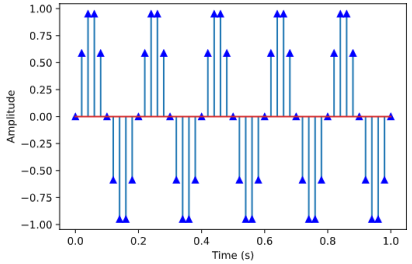
$$x(t) \rightarrow \boxed{\text{SAMPLING}} \rightarrow x_s(t) \rightarrow \boxed{\text{C/D}} \rightarrow x[n] \rightarrow \boxed{\text{QUANTIZATION}} \rightarrow x_q[n]$$

$x(t)$



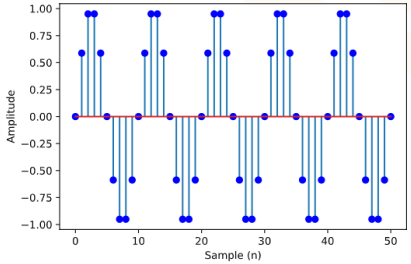
Continuous signal with amplitude values for all t possible

$x_s(t)$



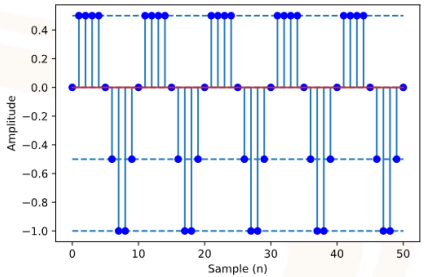
Sampled signal with values to $t = nT_s$. Amplitude represents the impulse area

$x[n]$



Discrete signal where time reference is transformed to samples

$x_q[n]$



Quantized signal, amplitude values are rounded to -1, -0.5, 0 or 0.5 (2 bits)

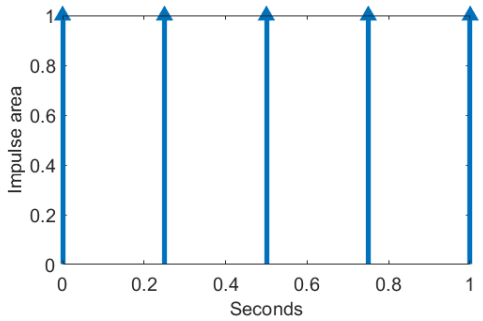
Modeling the sampling step

$$F_s = \frac{1}{T_s}$$

The sampling frequency defines the time interval (T_s) between the impulses

Increasing the F_s value (sample rate) decreases the T_s value, hence the number of impulses per second also increases

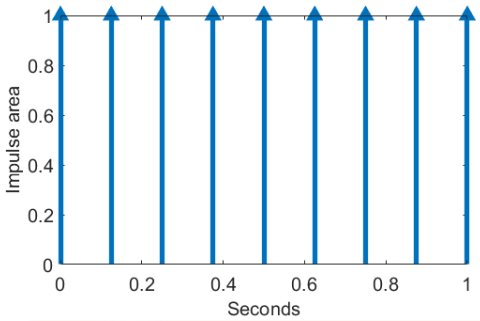
$F_s = 4$
$T_s = 1/4$



$F_s \times 2$

$T_s \div 2$

T_s defines the interval between impulses

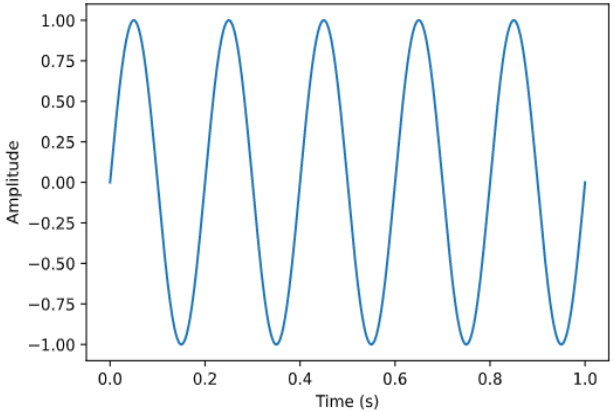


$F_s = 8$
$T_s = 1/8$

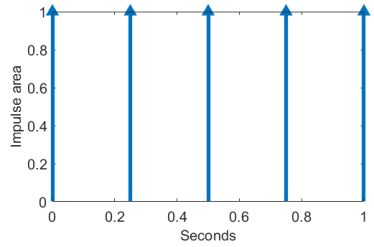
We call this an impulse train, given by $p(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$.

Modeling the sampling step

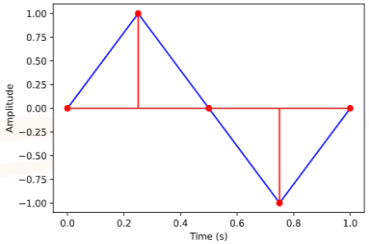
Higher sample rates provide better signal representation at the cost of higher prices to ADC

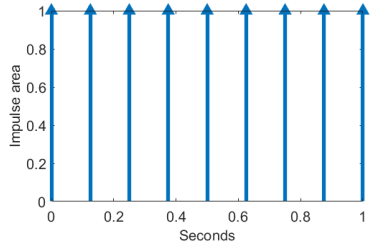


$y(t)=\sin(2\pi ft)$
$f = 5\text{Hz}$

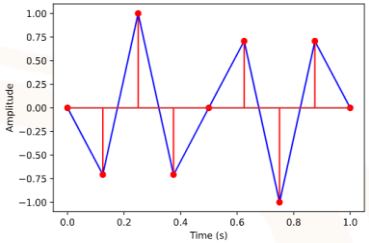


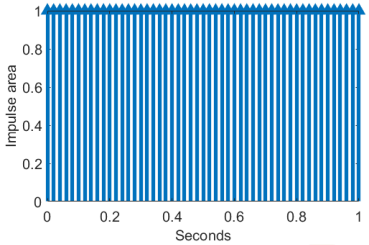
$F_s = 4\text{ Hz}$



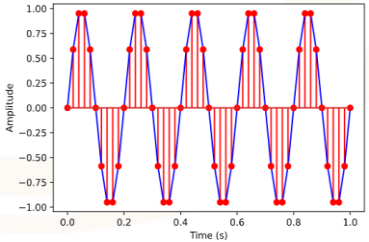


$F_s = 8\text{ Hz}$





$F_s = 50\text{ Hz}$

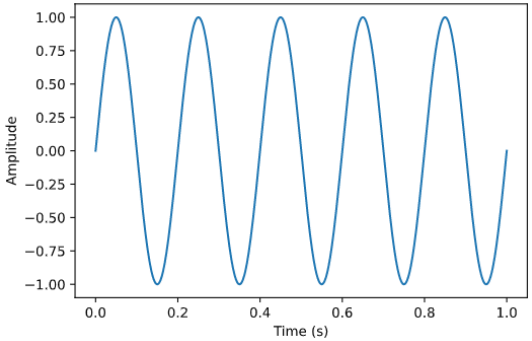


The analog sampled signal is $x_s(t) = x(t)p(t)$.

A/D sampling operation

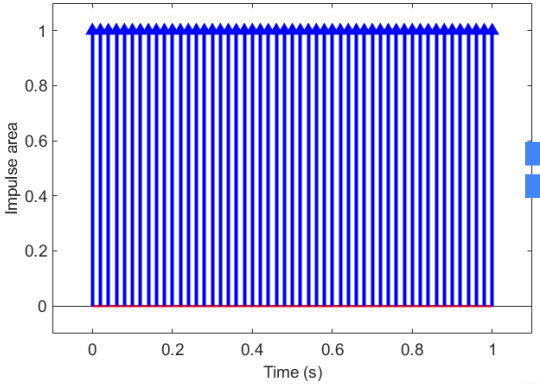
$$x(t) \rightarrow \boxed{\text{SAMPLING}} \rightarrow x_s(t)$$

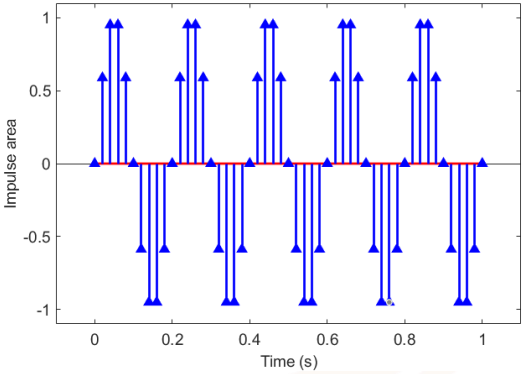
Sampling process uses the impulses to take samples from the signal in time steps defined by T_s



$y(t)=\sin(2\pi ft)$
$f = 5\text{Hz}$

$F_s = 50\text{ Hz}$



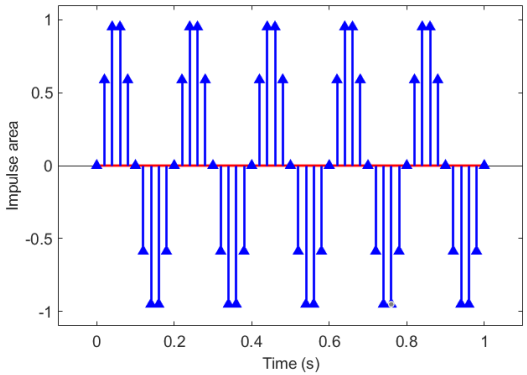


Dirac Delta
Amplitudes = area (plot)
Real amplitudes = infinity

C/D

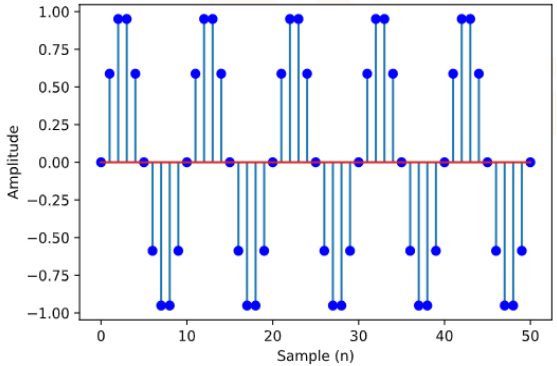
$$x_s(t) \rightarrow \boxed{\text{C/D}} \rightarrow x[n]$$

Transform the impulses in finite values determined by the area of sampled signal and convert from time to samples using a sampling frequency



$F_s = 50$ Hz

$$x[n] = x(t)|_{t=nT_s}$$



Now impulses have finite amplitudes (Kronecker delta)

Using Fs=50 Hz, each second has 50 samples. Therefore, to convert from time to samples we just need to multiply the time from sampled signal with Fs. For example, time instant t=0.6s becomes n=0.6*50 which is 30 in the discrete signal

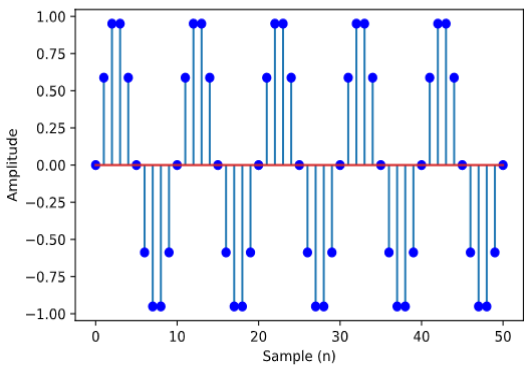
A/D quantization operation

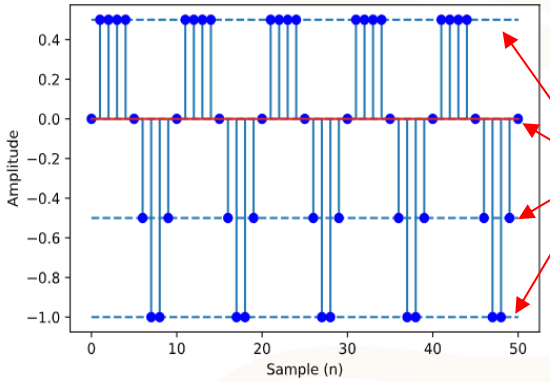
$$x[n] \rightarrow \boxed{\text{QUANTIZATION}} \rightarrow x_q[n]$$

A/D devices have finite number of bits to represent the amplitudes, so the "continuous" amplitude is discretized in one of the possible values quantized.

2 bits can represent 4 different values (quantization levels), for example:

$00 \Rightarrow -1$ $01 \Rightarrow -0.5$ $10 \Rightarrow 0$ $11 \Rightarrow 0.5$





Quantization levels

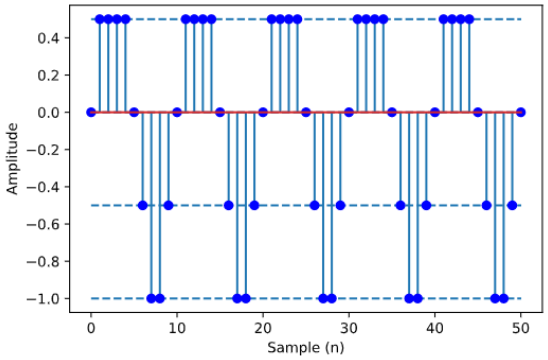
Note that the quantization process insert an error since the values are not represented identically to the discrete signal. For example, the signal amplitude with value 1 become 0.5 in the quantized signal since it is the nearest value among the options (-1, -0.5, 0 and 0.5).

Digital to analog (D/A) conversion = D/C + reconstruction (or interpolation)

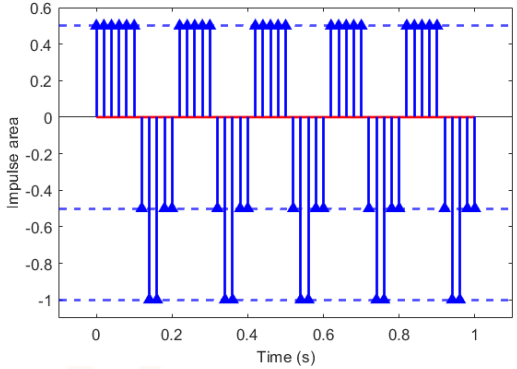
$$x_q[n] \rightarrow \boxed{\text{D/C}} \rightarrow x_{s,q}(t) \rightarrow \boxed{\text{RECONSTRUCTION}} \rightarrow x(t)$$

This stage converts a discrete-time quantized signal to continuous-time. It basically uses the sample frequency information to convert samples (digital signal) to time (analog signal). Then, it recreates the amplitudes in the points between Ts intervals.

First step represented by D/C is to convert samples (n) to a time scale (t):



$F_s = 50$ Hz



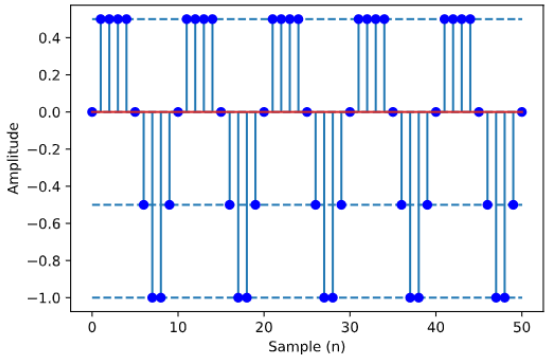
The samples (n) are converted to time (s) considering that time $t = nT_s = \frac{n}{F_s}$

D/C - Reconstruction / interpolation

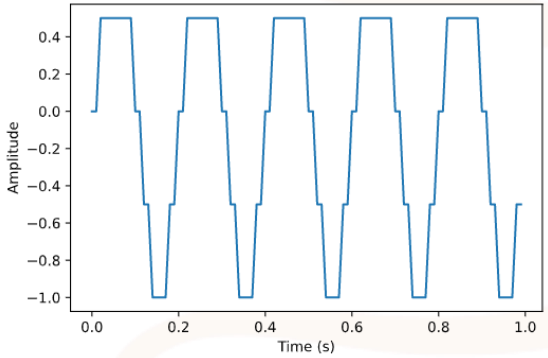
$$x_q[n] \rightarrow \boxed{\text{D/C}} \rightarrow x_{s,q}(t) \rightarrow \boxed{\text{RECONSTRUCTION}} \rightarrow x(t)$$

The second step is doing the reconstruction of the values in the space between samples and recover their amplitudes.

Many algorithms can reconstruct the signal with imperfections in relation to original signal. For example, the algorithm Zero-order Hold (ZOH) keeps the value of the previous sample until the next one.



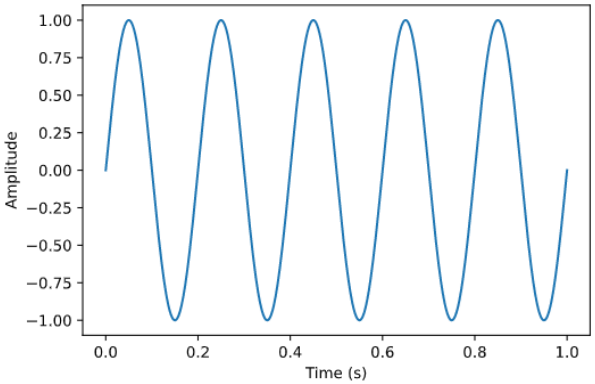
ZOH



ZOH keeps the sample value between the interval until the next sample

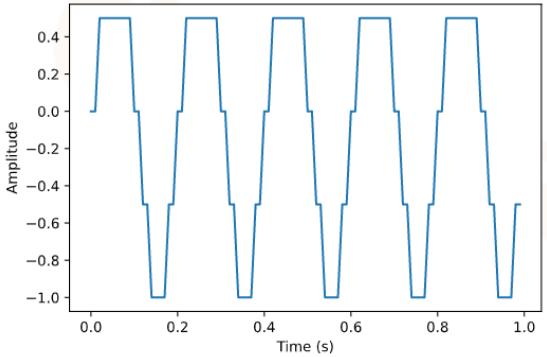
D/C - Reconstruction / interpolation

ZOH does not recuperate the original signal since it repeats the values between sampled times with the previous value



Original Signal

ZOH has distortions



Reconstructed Signal

ZOH can’t reconstruct original signal

Sampling theorem - Perfect reconstruction of signals

Sampling theorem defines: $F_s > 2f_{max}$

If the sampling theorem is obeyed, the original signal can be perfectly reconstructed from its discrete version using the following:

$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\mathrm{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

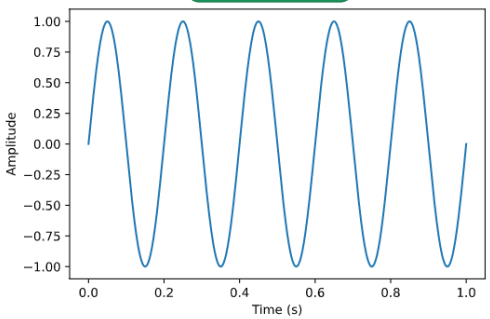
Try to understand the formula above as the x[n] signal amplitudes multiplying shifted and scaled sincs

Sampling theorem - Perfect reconstruction of signals

Perfect reconstruction:

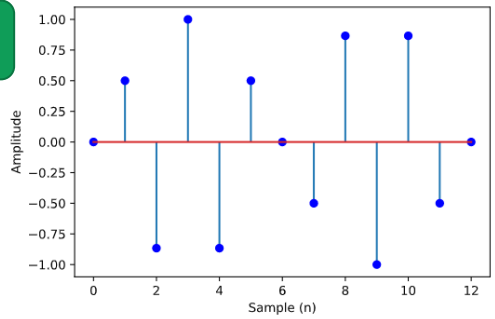
$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\mathrm{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

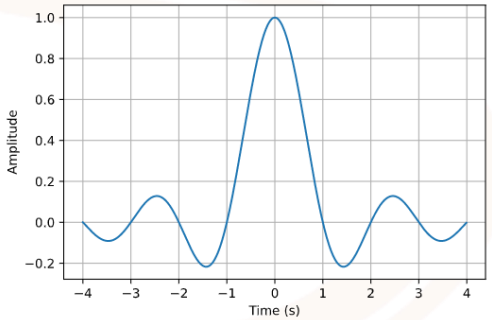
$x(t)$



$F_s = 12$ Hz

$x[n]$





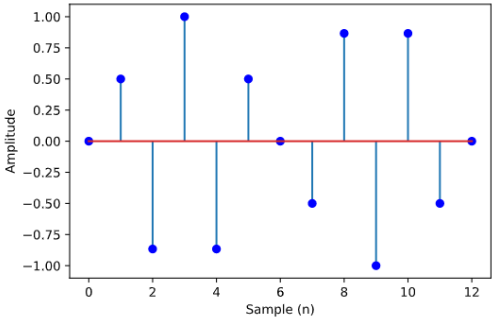
$y(t)=\sin(2\pi ft)$
$f = 5\text{Hz}$

Signal is sampled with a Fs which obeys the sampling theorem

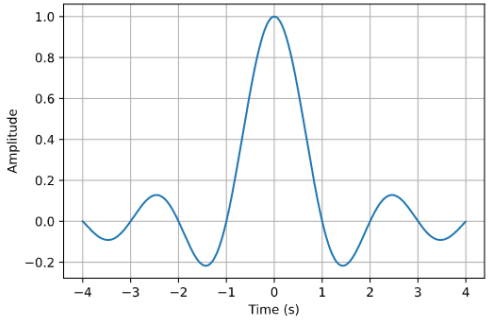
Sampling theorem - Perfect reconstruction of signals

There is one sinc located at each impulse and scaled by Ts

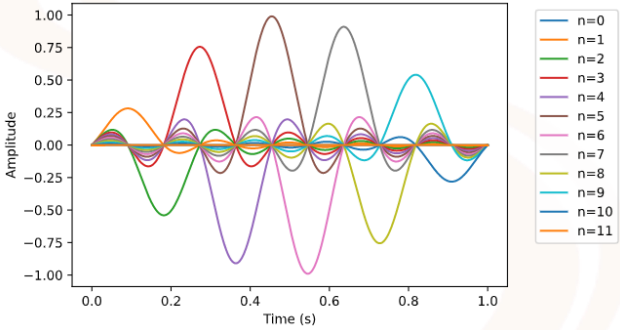
Each sinc carries part of the original signal, when sum all them, it should be obtained the original signal.



Discrete signal



Sinc function



All sinc functions multiplied by impulses from discrete signal

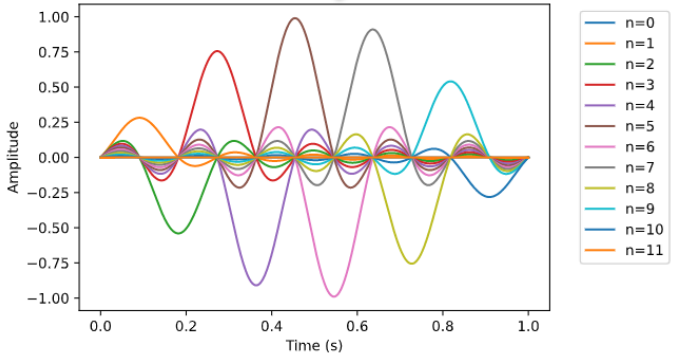
12 sinc functions compose the original signal

Sampling theorem - Perfect reconstruction of signals

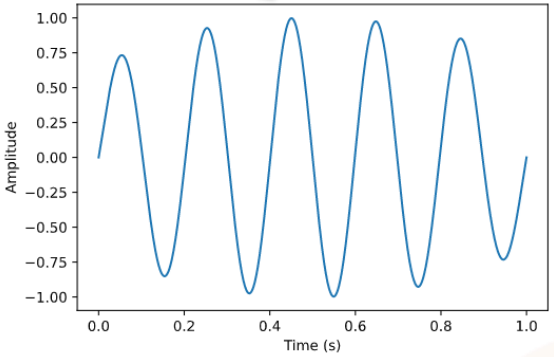
Summing all sinc curves we should obtain the original signal

All sinc functions were summed

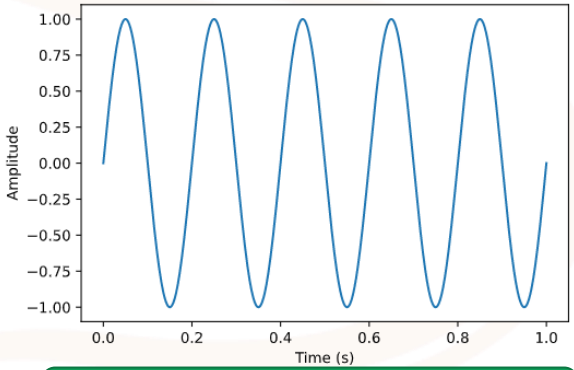
Reconstructed signal is different from original signal. Why?



Sinc functions



Reconstructed signal

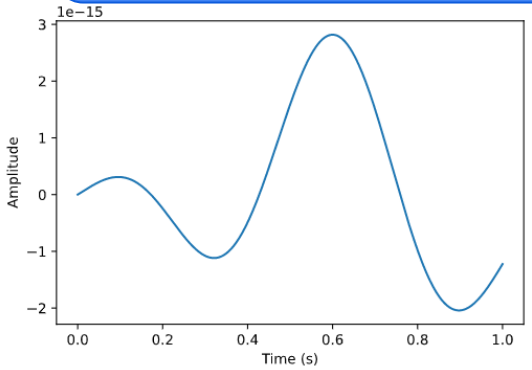


Original signal

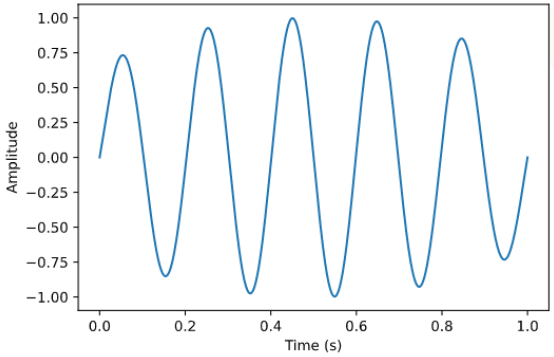
Sampling theorem - Perfect reconstruction of signals

Sinc function is an infinite signal which can not be perfectly represented by a computer. So, mathematically we could make a perfect reconstruction, but not using a computer.

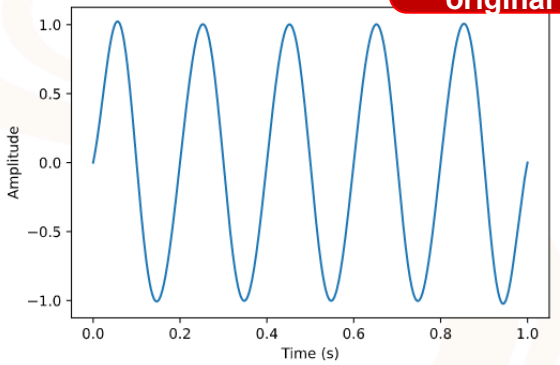
Seems close to the original



Reconstructed signal with Fs = 5 Hz



Reconstructed signal with Fs = 11 Hz



Reconstructed signal with Fs = 20 Hz

The sample rate can be increased to reach a higher accuracy to the reconstructed signal in relation to original

Quiz-Time!

A/D & D/A

Sampling Theorem

Quantization

Thank you!

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